

## 41.2 Complete cases and quick repairs

Confronted with holes, software does one of two things without being asked: it deletes the rows that have any, or it fills them and proceeds as though nothing had happened. Both are defensible in circumstances that can be stated precisely, and disastrous outside them. This section states them.

### 41.2.1 When deleting rows is safe

Write  $C_i = 1$  when unit  $i$ 's record is complete in every variable the analysis uses. A **complete-case analysis** applies the analysis to the subsample  $\{i : C_i = 1\}$ , and its behaviour is governed by one thing: the conditional distribution of  $\mathbf{d}_{(i)}$  given  $C_i = 1$ .

#### Proposition 41.2.1 — Validity of a complete-case analysis

Let the rows  $(\mathbf{d}_{(i)}, C_i)$  be independent and identically distributed, with  $\Pr(C_i = 1) > 0$ .

- (a) (*General condition.*) Let a procedure estimate a functional  $T$  consistently at every law in a class containing both  $f$ , the law of  $\mathbf{d}_{(i)}$ , and  $f_{C=1}$ , its conditional law given  $C_i = 1$ . Applied to the complete cases the procedure stays consistent for  $T(f)$  if and only if  $T(f_{C=1}) = T(f)$ . Under MCAR  $f_{C=1} = f$ , and every such functional is preserved.
- (b) (*Regression with selection on the covariates.*) Partition  $\mathbf{d}_{(i)} = (y_i, \mathbf{x}_{(i)})$  and suppose

$$\Pr(C_i = 1 \mid y_i, \mathbf{x}_{(i)}) = \Pr(C_i = 1 \mid \mathbf{x}_{(i)}) \quad \text{almost surely.} \quad (41.2.1)$$

Then the conditional law of  $y_i$  given  $\mathbf{x}_{(i)}$  is the same among the complete cases as in the population. So if a model specifies that conditional law correctly, the complete-case least squares estimate in a linear model is unbiased for  $\boldsymbol{\beta}$  conditionally on the selected covariates; and for a generalized linear model (Definition 34.2.1) the complete-case maximum likelihood estimate is consistent and asymptotically normal, with the usual model-based covariance of Theorem 34.5.1 computed from the complete cases alone, provided their number grows and their covariate second moments converge to a nonsingular limit.

- (c) (*Otherwise.*) If Eq. 41.2.1 fails, write  $\pi(y, \mathbf{x}) = \Pr(C = 1 \mid y, \mathbf{x})$ . The complete-case regression function is

$$E(y \mid \mathbf{x}, C = 1) = E(y \mid \mathbf{x}) + \frac{\text{Cov}\{y, \pi(y, \mathbf{x}) \mid \mathbf{x}\}}{E\{\pi(y, \mathbf{x}) \mid \mathbf{x}\}}, \quad (41.2.2)$$

so the complete cases follow the intended regression function exactly when  $\pi$  is conditionally uncorrelated with the response at every  $\mathbf{x}$ .

*Proof.*

- (a) The complete cases are an independent sample from  $f_{C=1}$ , and their number grows without bound since  $\Pr(C = 1) > 0$ . By the assumed consistency of the procedure at  $f_{C=1}$ , it converges to  $T(f_{C=1})$ , which is the target  $T(f)$  if and only if the two

agree. Under MCAR,  $C$  is independent of  $\mathbf{d}$ , hence  $f_{C=1} = f$ .

(b) By Bayes's rule,

$$f(y \mid \mathbf{x}, C = 1) = \frac{\Pr(C = 1 \mid y, \mathbf{x}) f(y \mid \mathbf{x})}{\Pr(C = 1 \mid \mathbf{x})}, \quad (41.2.3)$$

and under Eq. 41.2.1 the two selection probabilities cancel, leaving  $f(y \mid \mathbf{x})$ . The complete cases are therefore a sample from a joint law whose conditional of  $y$  given  $\mathbf{x}$  is the modelled one, with only the marginal law of  $\mathbf{x}$  distorted. Every likelihood built from  $f(y \mid \mathbf{x}; \boldsymbol{\beta})$  alone conditions on the covariates and is unaffected by that distortion.

(c) Multiply Eq. 41.2.3 by  $y$  and integrate:

$$\begin{aligned} E(y \mid \mathbf{x}, C = 1) &= \frac{E\{y \pi(y, \mathbf{x}) \mid \mathbf{x}\}}{E\{\pi(y, \mathbf{x}) \mid \mathbf{x}\}} \\ &= E(y \mid \mathbf{x}) + \frac{\text{Cov}\{y, \pi(y, \mathbf{x}) \mid \mathbf{x}\}}{E\{\pi(y, \mathbf{x}) \mid \mathbf{x}\}}, \end{aligned}$$

using  $E(y\pi) = E(y)E(\pi) + \text{Cov}(y, \pi)$  throughout at fixed  $\mathbf{x}$ .

■

Part (b) is the most useful positive result in the subject and is routinely overlooked. *Missingness may depend on the covariates as strongly as it likes, in any functional form, without harming a correctly specified regression.* The complete cases are a biased sample of units — their covariate distribution is wrong, so any marginal summary of them is wrong — and yet the fitted conditional mean is right. This is selection on a regressor, as in Chapter 25; the binary-response version is Exercise (B1).

**Warning.** Part (b) needs the conditional *model* to be right. If the true regression is curved and the fitted one is a straight line, selecting on  $\mathbf{x}$  changes which part of the covariate range dominates the fit, and the complete-case slope moves with it. Selection on the covariates is harmless for a correct model and can be very harmful for a wrong one.

### Example. (Four mechanisms, one regression)

Take  $y_i = 1 + 0.8x_i + 0.9w_i + \varepsilon_i$  with  $x_i, w_i, \varepsilon_i$  independent standard normal and  $n = 300$ , and lose the response under four mechanisms: completely at random; at random on  $x$ ; at random on  $x$  and  $w$ ; and not at random, on  $y$ .

**Table 41.2.1.** Mean complete-case estimate of  $\beta_1 = 0.8$  over 2000 replicates, for two analyses under four mechanisms.

| mechanism          | complete cases | fit $y \sim x$ | fit $y \sim x + w$ |
|--------------------|----------------|----------------|--------------------|
| MCAR               | 194            | 0.801          | 0.801              |
| MAR on $x$         | 181            | 0.804          | 0.802              |
| MAR on $x$ and $w$ | 178            | 0.637          | 0.800              |
| MNAR on $y$        | 178            | 0.593          | 0.669              |

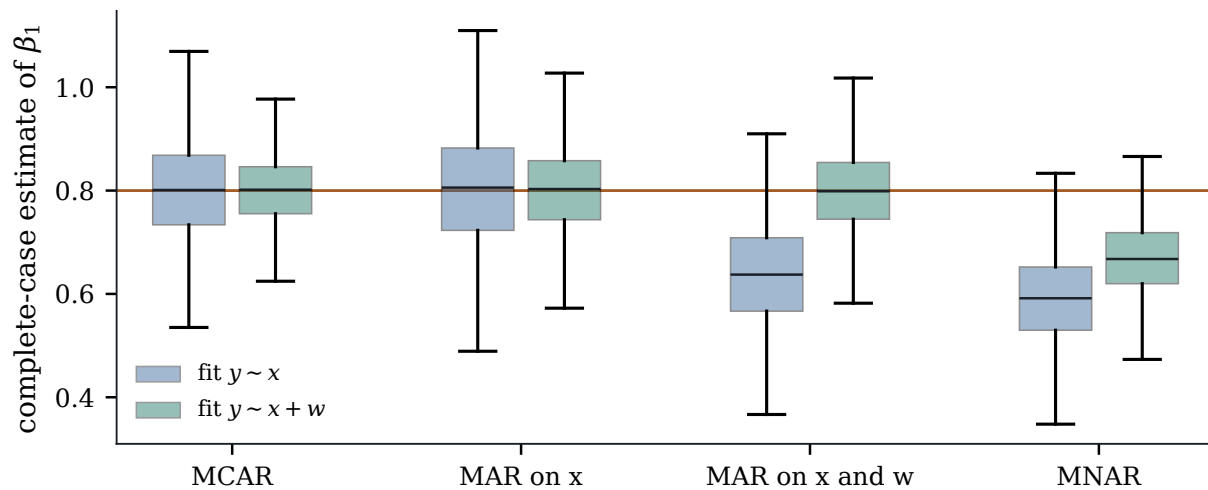


Figure 41.2.1: Complete-case estimates of  $\beta_1 = 0.8$  over 2000 replicates under four mechanisms, for the regression of  $y$  on  $x$  alone and on  $x$  and  $w$ . The horizontal line is the true value.

The first two rows are [Proposition 41.2.1\(b\)](#): whatever the missingness does with  $x$ , the regression of  $y$  on  $x$  is untouched. The third is the warning attached to it. The mechanism is MAR, using only recorded variables, yet the short regression omitting  $w$  is badly biased, because selecting on  $w$  selects on the part of  $y$  that the short model calls error; adding  $w$  repairs it completely. The fourth row cannot be repaired by any choice of regressors, because the selection is on  $y$  itself.

```
import numpy as np

rng = np.random.default_rng(4101)
n, reps = 300, 2000
beta = np.array([1.0, 0.8, 0.9])  # intercept, coefficient of x,
                                # of w

def one_run(mechanism, rng):
    x, w = rng.normal(size=n), rng.normal(size=n)
    y = beta[0] + beta[1] * x + beta[2] * w + rng.normal(size=n)
    if mechanism == "MCAR":
        eta = np.full(n, -0.6)
    elif mechanism == "MAR on x":
        eta = -0.6 + 1.6 * x  # depends on a regressor only
    elif mechanism == "MAR on x and w":
        eta = -0.6 + 1.2 * x + 1.5 * w  # depends on observed
    # variables only
    else:
        eta = -0.6 + 1.2 * (y - beta[0])  # MNAR
    # depends on the value itself
    missing = rng.uniform(size=n) < 1 / (1 + np.exp(-eta))
    obs = ~missing
    short = np.column_stack([np.ones(obs.sum()), x[obs]])  # y on x
    long = np.column_stack([np.ones(obs.sum()), x[obs], w[obs]])  # y on x and w
    b_short, *_ = np.linalg.lstsq(short, y[obs], rcond=None)
    b_long, *_ = np.linalg.lstsq(long, y[obs], rcond=None)
    return b_short[1], b_long[1], obs.sum()

names = ["MCAR", "MAR on x", "MAR on x and w", "MNAR"]
```

```

out = {m: np.array([one_run(m, rng) for _ in range(reps)]) for m in names}
for m in names:
    s, l, k = out[m].T
    print(f"{m:16s} complete cases {k.mean():5.0f}    y ~ x: {s.mean():.4f}"
          f"    y ~ x + w: {l.mean():.4f}")

```

[Run this code online](#)

*Idea.* Ask what the missingness depends on, then put those variables into the analysis. If everything the mechanism uses is in the model and the model is right, the complete cases are enough. If the mechanism uses the response, or a variable the model leaves out, they are not.

## 41.2.2 Available cases

A tempting middle course computes each summary from whichever records carry the variables it needs: a mean from everyone who reported that variable, a covariance from everyone who reported both. This **available-case** analysis, or pairwise deletion, uses more data, at the price of a covariance matrix whose entries come from different subsamples and need not be jointly realizable.

### Example. (A correlation matrix with a negative eigenvalue)

Twelve records on three variables, in three patterns of four:  $d_3$  absent in the first,  $d_2$  in the second,  $d_1$  in the third, with numbers chosen so that the available pairs give  $r_{12} = 0.9$ ,  $r_{13} = 0.9$  and  $r_{23} = -0.9$ . Each is an ordinary correlation on its own four points; together they are impossible, since by Proposition 2.4.2 the third must lie between  $0.81 - \sqrt{0.19 \times 0.19} = 0.62$  and  $0.81 + 0.19 = 1$ . The matrix has smallest eigenvalue  $-0.800$ , so any procedure needing it to be nonnegative definite — a Cholesky factor, a GLS weight matrix, a normal likelihood — fails, even though the matrix itself is perfectly invertible.

```

nan = np.nan
u = np.array([-1.5, -0.5, 0.5, 1.5])
d = np.array([0.5, -1.5, 1.5, -0.5])          # orthogonal to u, same length
pair = np.column_stack([u, 0.9 * u + np.sqrt(1 - 0.81) * d])      #
    correlation +0.9
flip = np.column_stack([u, -(0.9 * u + np.sqrt(1 - 0.81) * d)])    #
    correlation -0.9
D = np.vstack([
    np.column_stack([pair, np.full(4, nan)]),          # pattern A: z3 absent
    np.column_stack([pair[:, 0], np.full(4, nan), pair[:, 1]]),    # pattern B
    : z2 absent
    np.column_stack([np.full(4, nan), flip]),          # pattern C: z1 absent
])
R = np.ones((3, 3))
for i in range(3):
    for j in range(i + 1, 3):
        both = ~np.isnan(D[:, i]) & ~np.isnan(D[:, j])
        R[i, j] = R[j, i] = np.corrcoef(D[both, i], D[both, j])[0, 1]
print("available-case correlation matrix\n", R)
print("eigenvalues", np.round(np.linalg.eigvalsh(R), 3))

```

[Run this code online](#)

Under MCAR each entry is consistent and the difficulty is a finite-sample one; under MAR the entries are not consistent at all. Available-case analysis has uses in exploratory work and none in inference.

### 41.2.3 Single imputation

The other reflex is to fill the holes and carry on. **Mean imputation** replaces every missing entry of a variable by the mean of its recorded entries, leaving a pile of entries exactly at the sample mean and a spread that is too small.

#### Proposition 41.2.2 — Mean imputation deflates the variance

Let  $z_1, \dots, z_n$  be independent and identically distributed with finite variance  $\sigma^2$ , of which  $k$  are recorded and  $n - k$  missing, the mechanism being completely at random; let  $\tilde{z}_i$  equal  $z_i$  when recorded and  $\bar{z}_{\text{obs}}$  otherwise. Then the sample variance of the filled column satisfies

$$\mathbb{E}\left\{\frac{1}{n-1} \sum_{i=1}^n (\tilde{z}_i - \bar{\tilde{z}})^2 \mid k\right\} = \frac{k-1}{n-1} \sigma^2, \quad (41.2.4)$$

so the estimate of  $\sigma^2$  is too small by the factor  $(k-1)/(n-1) \approx k/n$ , the proportion recorded.

*Proof.* The filled column's mean is  $\bar{\tilde{z}} = \bar{z}_{\text{obs}}$ , because the  $n - k$  inserted values are all equal to it. The imputed entries therefore contribute nothing to  $\sum_i (\tilde{z}_i - \bar{\tilde{z}})^2$ , which reduces to  $\sum_{i \text{ obs}} (z_i - \bar{z}_{\text{obs}})^2$ . Given  $k$ , and under MCAR, that sum has expectation  $(k-1)\sigma^2$ . Dividing by  $n-1$  gives [Eq. 41.2.4](#). ■

With  $n = 300$  and 30% of values missing the factor is 0.699, and the simulation gives 0.700 as the average filled-in sample variance of a standard normal column. Covariances with fully recorded variables are deflated by the same mechanism, so correlations shrink towards zero and coefficients with them.

**Regression imputation** replaces a missing  $z_i$  by its predicted value from a regression on the recorded variables. It fixes the mean but not the spread, since the imputed points lie exactly on the fitted surface. In the same simulation, filling 30% of a variable whose correlation with the predictor is 0.6 raises the sample correlation to 0.667 and lowers the standard deviation to 0.899. Adding a draw from the estimated residual distribution — **stochastic regression imputation** — repairs both, and is the first honest imputation on this list. What it does not repair is the *uncertainty*: the analysis treats  $n$  values as measured when  $n - k$  were invented, and every standard error is too small. [Section 41.5](#) measures the damage — a nominal 95% interval covering 75.9% of the time in the example there — and repairs it.

*Warning.* Never treat a single set of imputed values as data. The estimate may be fine; the standard error, the  $t$  statistic and the interval are not, because the imputed values

are correlated with the imputation model's own estimates. The same warning appeared in [Section 18.6](#) for the filled-in plot of a designed experiment, where Theorem 18.6.1(e) gives the exact correction available in that special case.

## 41.2.4 Weighting

Survey statisticians treat unit nonresponse differently. If unit  $i$  is complete with probability  $\pi_i = \Pr(C_i = 1 \mid \mathbf{d}_{(i)})$ , then for any estimating function  $\mathbf{g}$  with  $E\{\mathbf{g}(\mathbf{d}; \boldsymbol{\beta})\} = \mathbf{0}$  at the true  $\boldsymbol{\beta}$ , the **inverse-probability weighted** equations

$$\sum_{i=1}^n \frac{C_i}{\pi_i} \mathbf{g}(\mathbf{d}_{(i)}; \boldsymbol{\beta}) = \mathbf{0} \quad (41.2.5)$$

are unbiased, because  $E(C_i/\pi_i \mid \mathbf{d}_{(i)}) = 1$ . Under MAR,  $\pi_i$  is a function of recorded variables and can be estimated, typically by a logistic regression of  $C_i$  on them, and the estimator is then consistent when that model is right. This is Horvitz and Thompson's (1952) idea transplanted from sampling to nonresponse, and its weakness is visible in the formula: a unit with a small estimated  $\pi_i$  receives a huge weight, so the estimator is least stable where the data are thinnest. Augmenting [Eq. 41.2.5](#) with a term built from a model for  $E(\mathbf{g} \mid \text{recorded})$  gives the **doubly robust** estimators of Robins, Rotnitzky and Zhao (1994), consistent if either model is right. Weighting is the natural language for unit nonresponse, where there is nothing to impute; likelihood and imputation are the language for item nonresponse, where there is.

## 41.2.5 Exercises

### 41.2.5.1 A. Check your understanding

#### Exercise A1.

Under the mechanism of [Proposition 41.2.1\(b\)](#), show that the complete-case sample mean of  $y$  is generally not consistent for  $E(y)$ , even though the complete-case regression coefficients are consistent. Reconcile the two statements.

*Solution.* Selection changes the law of  $\mathbf{x}$ , and  $E(y) = E\{E(y \mid \mathbf{x})\}$  averages the conditional mean over that law; the complete cases average it over the selected law instead. The regression coefficients describe  $E(y \mid \mathbf{x})$ , which selection leaves alone, so both statements hold at once: the conditional model is right and the marginal average of it is wrong.

### 41.2.5.2 B. Practice

#### Exercise B1.

In [Eq. 41.2.2](#) take  $y \mid \mathbf{x} \sim N(\mathbf{x}^\top \boldsymbol{\beta}, \sigma^2)$  and  $\pi(y, \mathbf{x}) = \Phi(a + by)$ . Show that

$$E(y \mid \mathbf{x}, C = 1) = \mathbf{x}^\top \boldsymbol{\beta} + \frac{b\sigma^2}{\sqrt{1 + b^2\sigma^2}} \frac{\phi(c)}{\Phi(c)}, \quad c = \frac{a + b\mathbf{x}^\top \boldsymbol{\beta}}{\sqrt{1 + b^2\sigma^2}},$$

and deduce that the complete-case regression function is nonlinear in  $\mathbf{x}$  whenever  $b \neq 0$ .

*Solution.* With  $y = \mathbf{x}^\top \boldsymbol{\beta} + \sigma e$ ,  $e$  standard normal,  $E\{\Phi(a + by)\} = \Phi(c)$  by the standard convolution identity, and  $E\{e \Phi(a + by)\} = b\sigma\phi(c)/\sqrt{1 + b^2\sigma^2}$ , obtained by writing  $\Phi(a + by) = \Pr(U \leq a + by)$  for an independent standard normal  $U$  and integrating by parts (equivalently, by Stein's identity  $E\{e h(e)\} = E\{h'(e)\}$ ). Dividing the second by the first and substituting into Eq. 41.2.2 gives the display. The added term depends on  $\mathbf{x}$  through  $c$ , and the ratio  $\phi/\Phi$  is not affine, so the complete-case mean is not linear in  $\mathbf{x}$ ; in particular no single shift of the intercept absorbs it.

### Exercise B2.

Verify Eq. 41.2.4 by simulation for  $n = 300$  and three missing fractions. Then show that the sample covariance of the filled column with a fully recorded variable  $w$  carries the *same* factor  $(k - 1)/(n - 1)$ , and explain why the fraction recorded  $k/n$  is only the limiting form of it.

*Solution.* The imputed entries satisfy  $\tilde{z}_i - \bar{\tilde{z}} = 0$ , so  $\sum_i (\tilde{z}_i - \bar{\tilde{z}})(w_i - \bar{w}) = \sum_{\text{obs}} (z_i - \bar{z}_{\text{obs}})w_i$ . Given  $k$  and under MCAR its expectation is  $k\sigma_{zw} - \sigma_{zw} = (k - 1)\sigma_{zw}$ , the second term coming from  $E(\bar{z}_{\text{obs}} \sum_{\text{obs}} w_i) = \sigma_{zw}$ . Dividing by  $n - 1$  gives  $\{(k - 1)/(n - 1)\}\sigma_{zw}$ , the factor of Eq. 41.2.4. The two candidates differ only through the divisors used for the sample moments, and  $(k - 1)/(n - 1) \rightarrow k/n$  as both grow:  $k/n$  is the population-level statement,  $(k - 1)/(n - 1)$  the exact finite-sample one, and there is no second phenomenon hiding in the gap.

### Exercise B3.

Prove that Eq. 41.2.5 is an unbiased estimating equation when  $\pi_i > 0$  for every unit, and show by example that the conclusion fails if  $\pi_i = 0$  for a set of covariate values with positive probability. What does that failure correspond to in Proposition 41.2.1?

#### 41.2.5.3 C. Going deeper

### Exercise C1.

Let  $E(y | x) = \beta_0 + \beta_1 x + \beta_2 x^2$  with  $\beta_2 \neq 0$ , and let the analyst fit a straight line. Show that even under Eq. 41.2.1 the complete-case slope generally differs from the full-data slope, and identify the functional of the covariate distribution that the straight-line slope estimates (see Theorem 2.5.2 and Theorem 6.11.1). Under what selection would the two agree?

*Solution.* The straight-line least squares coefficient converges to the best linear predictor coefficient  $\text{Cov}(y, x)/\text{Var}(x)$  computed under whatever law  $x$  has. With  $E(y | x) = \beta_0 + \beta_1 x + \beta_2 x^2$  this is  $\beta_1 + \beta_2 \text{Cov}(x^2, x)/\text{Var}(x)$ , which depends on the third and second central moments of  $x$ . Selection changes those moments, hence the limit, unless it leaves  $\text{Cov}(x^2, x)/\text{Var}(x)$  unchanged — for instance under MCAR, or under a selection probability that is an even function of  $x$  about its mean when the law of  $x$  is symmetric.